

Product Support Bulletin

PSB #: PSB-A11-03 Upgrade N2 feeder door latch.
System: H9000
Date: February 2011

Overview

ECN008842 was released to upgrade the existing feeder door latch to a new mechanism in order to improve performance and reliability.

Systems Affected

All Neptune 2 systems shipped before May 2011 could be affected.

Compatibility

No issues

Impact

Feeder door will be open for 10 minutes in order to execute this PSB

Technical Description

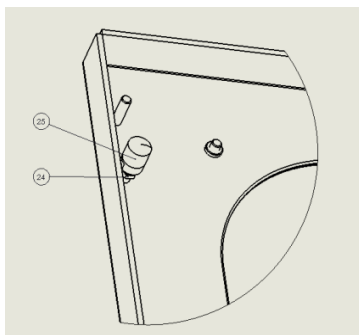
This is an upgrade to the feeder door elevator interlock. These will improve reliability and safety. It will replace the spike with a rubber bumper. It will also add on a slider assembly to the interlock that will improve performance.

Tool List

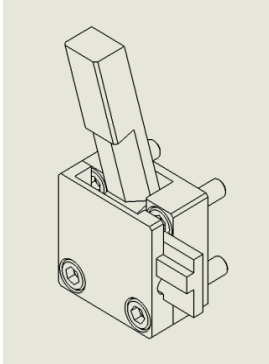
Part Number	Description	Q
470-471-09	SCRW CP SKT HD M6X1.0P/30MM SS	2
601-881-01	SPRING COMP 7/32ODx1"L MUSIC WR	1
612-434-00	ME FEEDER DOOR ELEVATOR INTERLOCK SLIDER HOUSING	1
612-436-00	ME FEEDER DOOR ELEVATOR INTERLOCK SLIDER	1
606-631-05	PUSHER M5x12DI Ax16MML, SS-POLYURETHN	1

Installation Instruction

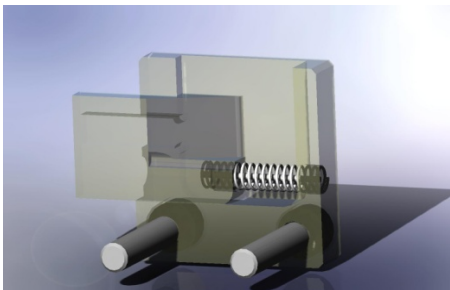
1. Move the “raise tote” plate up and out of the way so that door can open freely.
2. Open feeder door (605-803-00) by loosening set screw (M4 Wrench) and removing screws (M3 and M4 Allen Wrenches) that hold the door closed.
3. Replace the spike assembly in Assembly BOM 605-803-00 with the rubber bumper (606-631-05). Remove spike and hardware attaching it to the door.
 - o Remove M5 Nuts (470-453-05) using M5 hex nut driver. One will be reused for the rubber bumper. (606-631-05)
 - o Remove two Lock Washers (470-660-01)
 - o Remove two Washers (470-501-03)
 - o Remove Spike (604-891-00, 604-892-00, 470-482-05)
4. Bumper and two nuts added to bottom stud.
 - o Screw on one of the nuts removed from spike (470-453-05) to bottom stud.
 - o Screw on rubber bumper (606-631-05) to bottom stud.
 - o Use nut to jam bumper in place at desired location after interlock installed.
 - You will need to tighten the bumper against the nut. You can use a 10mm wrench on the bumper flats, and a 8mm wrench on the M5 nut. Tighten the bumper clockwise and the nut counter-clockwise to a maximum torque of 29 in lbs to jam the bumper in place.



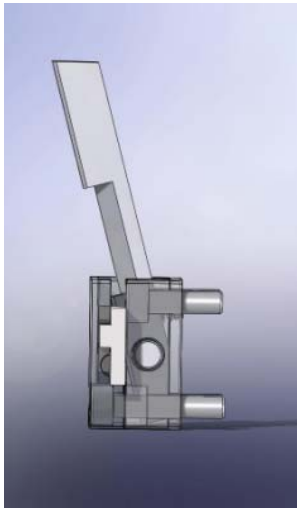
5. *Interlock add-on installed at the same time per BOM Assembly 605-965-00.*



- *Remove two bottom screws (470-471-07) on latch block (604-889-00) with M5 Wrench.*
- *Assemble interlock add-on per Assembly Drawing 605-965-00*
 - *Put spring (601-881-01) in hole on the slider (612-436-00).*
 - *Put slider into interlock housing (612-434-00) with spring sitting in hole on housing. The slider is held in the housing by the spring force created when the spring is put in the concentric holes of the housing and slider. (Picture Below)*
 - *Put two screws (470-471-09) through holes in housing per drawing.*



- *With fingers holding slider from coming out, install interlock upgrade onto existing latch block. (As shown below)*
- *Tighten two bottom screws to a **maximum** torque of 29 in lbs with a M5 wrench*



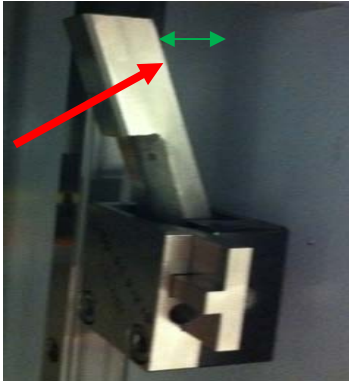
6. *Test the slider mechanism. Make sure slider returns when pushed and that it acts on the latch.*



Latch is parallel to feeder wall when door is closed and interlock engaged

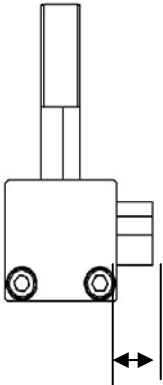
7. *Test to ensure latch on interlock is at an angle to the feeder wall when the door is open. This latch will need to hang off the wall at*

an angle to prevent the elevator from lowering when the door is open. The edge highlighted in the picture below should be at least 20 mm from the feeder wall. (See Picture Below)



When door open and latch disengaged the latch hangs free to stop the elevator from lowering

- 8. Next check that the latch becomes parallel to the wall when the door is closed. You may need to adjust the bumper on the stud so that it sticks out enough to push in the slider and makes the latch vertical. However, also adjust so that the bumper doesn't stick out too much so that the feeder door can't close flush. You may need to remove the feeder window to check the interlock operation.*



The distance here should be less than 2mm when the door is closed and the bumper is engaging.

This Page Intentionally Left Blank